Project 1 (30%)
CIT1C18 - Computational Thinking

Introduction

This individual assignment aims to design a flowchart for a problem within the confines of the Temasek Polytechnic. Students are reminded to read through the entire assignment specifications to understand the complete requirements.

It is the start of a new semester, and the new intake of Year 1 students are new to the campus and not familiar with the locations around campus.

Your task is to design the program logic using a flowchart that will help the students navigate to key locations in Temasek Polytechnic while considering real-world factors that affect movement, such as but not limited to:

- Weather Conditions:
  If it is raining, students might prefer covered walkways or indoor routes instead of open pathways. On good weather, the students might prefer the shortest route.

- Crowd Level:
  During peak hours (e.g., lunchtime), certain walkways, canteens, and entry points might be congested.

Learning outcomes:
By the end of this assignment, you should be able to:
- Use necessary tools, such as flowchart, to solve problems
- Describe and justify the problem-solving process

Note: The use of artificial intelligence tools, including but not limited to ChatGPT, is strictly prohibited for this project.

Task 1 – Location Selection and Visual Documentation (4%)

Walk through Temasek Polytechnic and identify THREE (3) locations that can be used as part of your problem.

Location 1 (Start Location):

Location 1 will be your starting location.
Choose one location from the options below as your start location.

- Lift Lobby A (Block 3, Level 3)
- Lift Lobby B (Block 7, Level 3)

Location 2:
Location 2 will be fixed: The **Little Skool House** (kindergarten) in the School of Humanities & Social Sciences (HSS)**.**

Location 3:
Location 3 **must be outside of the School of Informatics & IT premises**.
Choose one location from the options below.

- McDonald's
- School of Design Canteen (The Design Pad)
- School of Engineering Canteen (The Short Circuit)
- Temasek Convention Centre
- Swimming Complex
- Digital Fabrication & Additive Manufacturing Centre

Take a photo of each location and insert the **THREE (3) photos** in **Section A** of your report.

Important: The photos must be **clear and visually helpful**. They should display enough of the environment to help a user unfamiliar with the area understand their location and orientation.

Task 2 – Flowchart Design (16%)

Imagine that you need to code out a program that will guide students from the start location to the **2 (TWO)** selected locations in Task 1. (Example program below).

The program should minimally meet the following requirements:

- Display the start location.
- Prompt the user to input the destination location.
- Prompt the user to input details for real-world factors.
- Display the directions to the destination location.
- Run continuously until the user chooses to exit.

Example Program:

```
You are at Lift Lobby B (Block 7, Level 3)
Where do you want to go? (Enter: 1, 2 or 3)
1) Little Skool House
2) Mcdonald's
3) Exit
1

Is it a peak period right now? (Yes/No)no

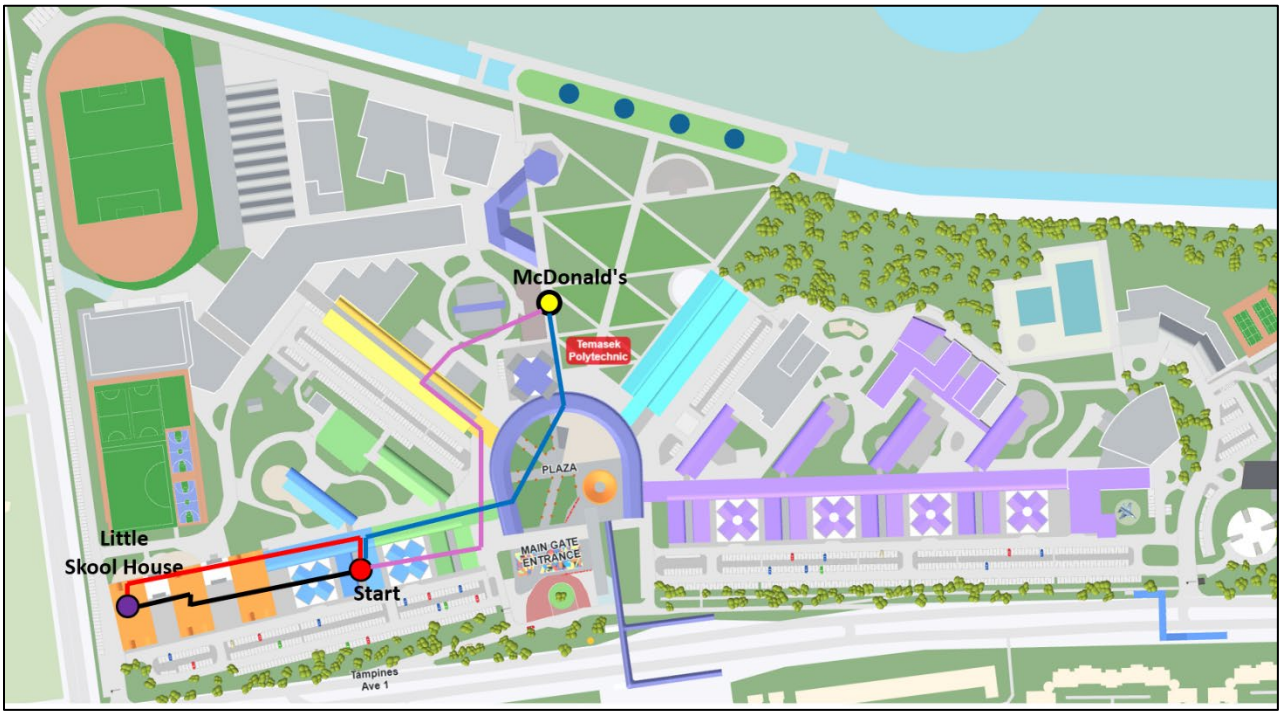
You have selected option: 1

Here are the directions to Little Skool House:
Turn to face Block 1A.
Walk straight towards Block 1A.
Walk straight towards Centre for Applie Behavioural Sciences.
Take [2] level of lift down
Turn to face Little Skool House.
Walk straight towards Little Skool House
You will have arrive at the destination

You are at Lift Lobby B (Block 7, Level 3)
Where do you want to go? (Enter: 1, 2 or 3)
1) Little Skool House
2) Mcdonald's
3) Exit
```

Based on your selected locations from Task 1:

1) Take a screenshot of the TP Campus Map and plot out the routes to the locations.
   (TP Campus Map - https://www.tp.edu.sg/about-tp/tp-campus-map.html)
   (Sample of plotted routes below)



2) **Create a flowchart** using the tool taught in class to illustrate the program.

   You must **limit your flowchart to no more than 60 blocks** (decisions, actions, and input/output blocks combined), **15 decision blocks** and **5 input blocks**.

   Here are the statements that the program prints to guide the user to the destination:

   - Walk straight towards [enter end point]
   - Turn to face [enter location]
   - Climb [x] level of the stairs up
   - Climb [x] level of the stairs down
   - Take [x] level of the lift up
   - Take [x] level of the lift down

   Your instructions should use nearby landmarks or recognisable points along the way to guide the user to the final destination. Avoid jumping directly to the final destination.

Insert the **map** and **flowchart** in **Section B** of your report.
(Can be a screenshot of the flowchart. Ensure that the flowchart is clearly visible.)

Task 3 –Route Justification and Evaluation (10%)

After designing your flowchart, using the routes that you had mapped out to **Location 3** and explain the following:

- Why did you choose these routes to reach Location 3 instead of other possible options. In your explanation, describe the different routes you considered, including the one you initially planned to take. Then, discuss how your choice was influenced or changed after considering real-world factors.

- What made you decide that these were the most suitable routes? What factors did you consider, and support your decision with data.

Your justification is not limited to just written words; you may include any form of data to support your decision.
You have a **500-word** limit.

In **Section C** of the report, include the explanation for your choice of route and any relevant data that may help clarify your decision-making process.

Submission
You are to create your Project 1 Word document report using the template ('COMT_AY2526_Project_1_Template.docx'). Submit your document into LMS under the Assessment folder by **Week 6, Thursday, 29th May 2025, at 2359hrs (11.59 pm).**
You should name your document according to the following format:

StudentID_YourName_COMT_AY2526_Project_1.docx

e.g.: 1234567D_Teo_Chu_Chu_COMT_AY2526_Project_1.docx

Penalty for Late Submission

late and < 1 day : 10% deduction from absolute mark given for the assignment
late >= 1 and < 2 days : 20% deduction from absolute mark
late >= 2 days : No marks awarded

Note that "day" includes **non-working days** (Sat, Sun and public holidays).

Project 1– Grading Criteria

The grading criteria for **Project 1 (30%)** will be based on the following:

Criteria	In Context	Performance Level				
		A (>=80%)	B (>=70%)	C (>=60%)	D (>=50%)	F (<50%)
		Excellent	Good	Average	Pass	Below Standard
Location Selection and Visual Documentation (4%)	Navigational Usefulness (4%) - Clarity - Usefulness of photos	Photos are clear, well-composed, and immediately help to recognise the location. Key features are intentionally shown to support orientation.	Photos clearly show the location and include some recognisable details, though could be better composed or include more helpful elements for full clarity	Photos show the location adequately, but may require some prior knowledge or assumptions to identify the place.	Photos capture the basic setting of the locations but lack clarity.	Non-submission Photos are unclear or unusable. It lacks sufficient information to support any understanding of the location.

Flowchart Design (16%)	Logic (10%) - Correctness - Completeness - Flowchart Optimization	The flowchart is logical, correct, and efficient, with no missing steps or errors in the sequence. Flowchart design fully meets the program requirements. Flowchart accounts for multiple real-world factors that affect movement and adjusts the route accordingly. All routes are optimised. The flowchart contains no unnecessary repeated steps. While there may be some necessary repeated steps to maintain clarity or ensure the process is followed correctly, the	The flowchart is logical and correct. Flowchart design fully meets the program requirements. Flowchart accounts for multiple real-world factors that affect movement, and adjusts the route accordingly. The flowchart contains some unnecessary repeated steps that do not complicate the overall solution.	The flowchart is mostly logical and correct. Flowchart design meets most of the program requirements. Flowchart includes primary route to the destination, taking into account one real-world scenario. The flowchart contains some repeated steps, which affect the clarity of the solution.	The flowchart is somewhat logical and correct. Flowchart design meets some of the program requirements. Flowchart includes primary route to the destination. The flowchart contains many repeated steps, complicating the solution unnecessarily.	Non-submission Incomplete flowchart Flowchart includes primary route to the destination, but jumps directly to the destination.

		overall solution is clear and concise.				
	Flow Chart Conventions (3%)	All standard flowchart symbols are used correctly, and the flow is clear and easy to follow. The layout is well-organized, and the symbols are correctly applied throughout, ensuring clarity and structure.	All standard flowchart symbols are correctly used, with a clear and logical flow that is mostly easy to follow. The layout is organised, with some room for improvement in terms of clarity.	Most of the standard flowchart symbols are correctly used. The layout lacks coherence, making the flowchart hard to follow.	Some of the standard flowchart symbols are correctly used. Poor organisation and/or unclear connection is present, making the flowchart difficult to follow	The flowchart does not use appropriate symbols, and the layout is disorganised. It is difficult to understand or completely unclear.
	Constraints (3%) - Keeping to the constraints and limitations specified	The flowchart strictly adheres to **all** constraints (no more than 60 blocks – Decisions, actions, and input/output blocks combined, 15 decision blocks and 5 input blocks).	The flowchart strictly adheres to all constraints (no more than 60 blocks – Decisions, actions, and input/output blocks combined, 15 decision blocks and 5 input blocks).	The flowchart adheres to at least two constraints (no more than 60 blocks – Decisions, actions, and input/output blocks combined, 15 decision blocks and 5 input blocks).	The flowchart adheres to at least one of the constraints (no more than 60 blocks – Decisions, actions, and input/output blocks combined, 15 decision blocks and 5 input blocks).	The flowchart shows little to no consideration of the constraints.

		Flowchart directions are clear and fully make use of the specified directional instructions listed in task 2.				

Route Justification and Evaluation (10%)	Route Justification and Evaluation (10%) - Clarity of justification - Thoroughness of Evaluation - Evidence of research	Clear, logical, and well-supported justification for the choice of routes, with strong reasoning for why these specific routes were chosen over possible alternatives. Thorough and detailed evaluation of all relevant factors showing a deep understanding of how each one impacts the route decisions. All research evidence presented strongly supports the justification of the route choices. The research data are highly relevant, well integrated and strongly reinforce the evaluation.	Clear and reasonable justification for the choice of routes, but more depth or detail could enhance the explanation. Solid evaluation of key factors but may lack some depth or overlook one or two aspects of the route choices. Multiple pieces of research evidence are provided, with proper justification for how they support the route choices. The research data are relevant and reinforce the evaluation.	Some justification for the choice of routes is provided, but the reasoning lacks clarity or depth, and further exploration of key aspects. The evaluation touches on some key factors but misses critical aspects, and the analysis lacks depth. Relevant research evidence is used and generally supports the route choices, though there may be minor gaps in explanation or integration into the overall evaluation.	Vague explanation provided of why the routes were chosen. The evaluation touches on a few relevant factors but is not explored in sufficient detail. Research evidence is present but only partially supports the route choices, with limited integration into the overall justification.	No clear justification is provided, or reasoning is completely absent. No evaluation of key factors. The analysis is missing or entirely superficial. There is limited research evidence to support the justification, and the explanation is vague or superficial.